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Exciton dressing by extreme nonlinear magnons in a layered semiconductor

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Collective excitations presenting nonlinear dynamics are fundamental phenomena with broad applications. A prime example is nonlinear optics, where diverse frequency-mixing processes are central to communication and attosecond science, and extreme (>sixth-order) harmonic generation provides broad wavelength conversion. Leveraging recent progress in van der Waals magnetic semiconductors, we demonstrate nonlinear optomagnonic coupling. In the layered antiferromagnetic semiconductor CrSBr, we observe exciton states dressed by up to 20 harmonics of magnons, resulting from their extreme nonlinearities. We also create tunable optical sidebands via sum- and difference-frequency generation between two optically bright magnon modes under symmetry-breaking magnetic fields. Moreover, we can tune the observed difference-frequency generation mode into resonance with one of the fundamental magnons, which results in parametric amplification of magnons. Our findings realize the modulation of the optical-frequency exciton with the extreme nonlinearity of magnons at microwave frequencies, which could find applications in magnonics and hybrid quantum systems, and provide a method for optomagnonic neuromorphic computing devices.

Magnonics is an emerging frontier that enables energy-efficient information processing from spin waves[1]. It has gained considerable interest[2–5] due to the efficient coupling of magnons with disparate excitations, such as superconducting qubits, microwave photons and phonons, making magnons suitable quantum transducers[6]. Advancements in two-dimensional (2D) semiconducting magnets have provided a route to couple magnons to excitons[7,8], while higher-order nonlinear excitations of quasiparticles in condensed-matter systems have led to controlled manipulation of materials properties such as lattice control via nonlinear phononics[9]. In magnets, nonlinearities are crucial for realization of magnon logic[10]. Thus, nonlinear magnonics[4,11–16] aims to combine the information-carrying capacity of magnons with interactions and controls from strong nonlinearities. An unprecedented opportunity arises from the exciton–magnon coupling in the 2D magnet CrSBr, as it may allow the mutual sensing or control afforded by nonlinear magnonics across many orders of magnitude in energy.

In CrSBr, a direct-gap 2D van der Waals semiconductor, intralayer and interlayer magnetic coupling are ferromagnetic (FM) and antiferromagnetic (AFM), respectively, with spins aligned along the crystal *b* axis at equilibrium[17,18]. Figure 1a shows the crystal structure and spin orientations at two distinct phases of the optical magnon precession, illustrated as arrows and separated by a dashed line. The magnon wavelength spans many unit cells and neighbouring spins have nearly identical orientations, so the vertical dashed line in Fig. 1a can be thought of as a translation to a distant point in the crystal. A recent report has demonstrated how the localization of the exciton wavefunction in

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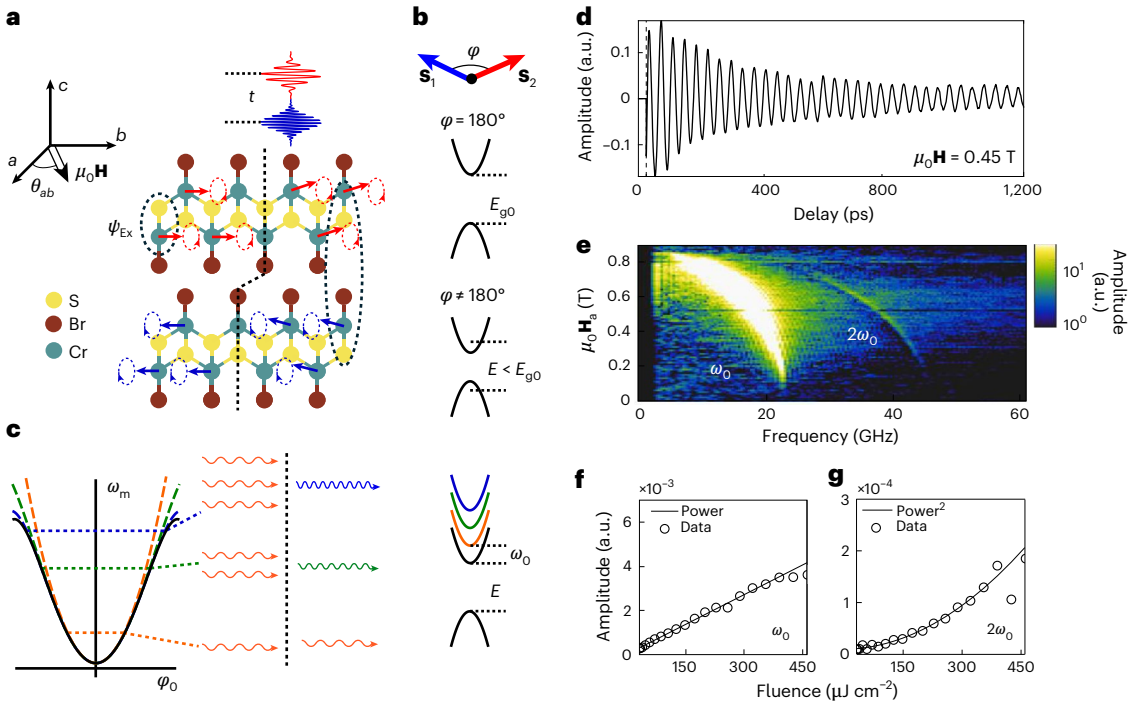

Fig. 1 | Coupling of excitons and nonlinear magnons in CrSBr. a, Crystal structure of CrSBr with two distinct spin orientations corresponding to distinct phases in the optical magnon precession. Black dashed ovals illustrate the delocalization of ψ_{Ex} at two different points in the spin wave where pump and probe pulses excite and measure the system at some delay, *t*. **b**, Schematic illustration of the exciton dependence on interlayer spin alignment in CrSBr, where E_{g0} is the exciton energy in the layered AFM state. **c**, Cartoon showing the nonlinear dependence of ω_m on φ_0 and its effect on the exciton in the perturbative regime. Dashed curves show approximations of varying orders. Horizontal dotted lines show regions where the behaviour of the system is described by the various approximations and are connected to interaction diagrams for harmonic generation processes. The right-hand cartoon depicts the resulting magnon-dressed exciton states. **d**, Transient optical reflectivity trace from thin CrSBr at μ_0**H** ≈ 0.45 T. a.u., arbitrary units. **e**, Magnon spectrum obtained from the FFT of the transient data. **f**,**g**, Power dependence of the FFT amplitudes of the fundamental (ω_0, **f**) and second-harmonic ($2\omega_0$, **g**) magnon modes measured at μ_0**H** ≈ 0.35 T. The magnetic field is applied along the crystal *a* axis for the data in **d**–**g**.

CrSBr, ψ_{Ex} in Fig. 1a, depends on its magnetic configuration[19], leading to an exciton energy shift proportional to the relative orientation of the spins (angle φ in Fig. 1b), that is, $\mathbf{S}_1 \cdot \mathbf{S}_2$, where $\mathbf{S}_i$ is the magnetization in an individual layer. The strong coupling between the exciton and interlayer spin alignment provides direct optical access to the magnons, where optical pulses can generate and detect the magnons through modulation of exciton properties in pump–probe experiments. This has led to reports of coherent magnon transport[7,20], hybridization of magnon modes[8] and magnon-coupled exciton-polaritons[21].

In this work, we report the observation of a series of optical sidebands generated via exciton dressing by nonlinear magnon modes in CrSBr. These nonlinear processes include high-harmonic generation (HHG), sum-frequency generation (SFG), difference-frequency generation (DFG) and parametric amplification. We show that we can tune the nonlinear response by breaking system symmetry to control the emergence of new magnon modes or amplify fundamental modes. We also show that we can reach the extreme non-perturbative nonlinear regime and access over 20th-harmonic generation in our measurements, where the term 'extreme' is borrowed from nonlinear optics and indicates higher than sixth-order nonlinearity[22,23]. Optical generation of high-density magnon populations with ultrafast pulses allows us to generalize the unprecedented optomagnonic coupling in CrSBr to the extreme nonlinear magnon regime.

Perturbative magnon generation

To measure nonlinear magnonic processes, we performed transient optical reflectivity measurements near the optical gap of 1.4 eV on exfoliated CrSBr as described in previous reports[7,8,21]. The experimental temperature of 35 K is far below the Néel temperature of ~132 K (ref. 24). Additional details can be found in Methods. Figure 1c represents the nonlinear dependence of the magnon frequency, ω_m, on the initial displacement of the spins from equilibrium, φ_0. With small spin displacements, the system is well described by a simple harmonic oscillator. At larger spin displacements, higher-order processes will appear and produce optical sidebands by modulating the exciton states (Fig. 1c, right panel).

Figure 1d shows a typical transient optical reflectivity trace with applied magnetic field μ_0**H** ≈ 0.45 T along the *a* axis. Non-oscillatory signal was removed from these data to emphasize the oscillations from the exciton–magnon coupling (raw data in Extended Data Fig. 1). The apparent signal decay is due to magnon propagation away from the measurement spot, rather than magnon dephasing[20]. Figure 1e presents the magnon spectrum, obtained through fast Fourier transformation (FFT) of the signal (*c*-axis field dependence in Extended Data Fig. 2). The mode spanning from ω_0 ≈ 5 GHz to ~30 GHz is the fundamental optical magnon sideband, as in previous reports[8,25,26]. The new observation is the emergence of a sideband at $2\omega_0$ with a related field dependence to the optical mode, signifying second-harmonic generation (SHG) of the fundamental magnon mode.

To support the SHG assignment, we perform fluence-dependent measurements. Figure 1f,g shows the FFT amplitudes of the ω_0 (Fig. 1f) and $2\omega_0$ (Fig. 1g) modes under increasing pump fluence. The amplitude of the ω_0 mode displays a linear power dependence until the magnon signal saturates at ~450 μJ cm^{-2}, while the amplitude of the $2\omega_0$ mode presents clear quadratic dependence up to the same saturation fluence. Since the bosonic magnons do not saturate due to Pauli blocking, the saturation is probably due to local laser heating. For the remaining results, we keep the pump fluence below 300 μJ cm^{-2} to minimize

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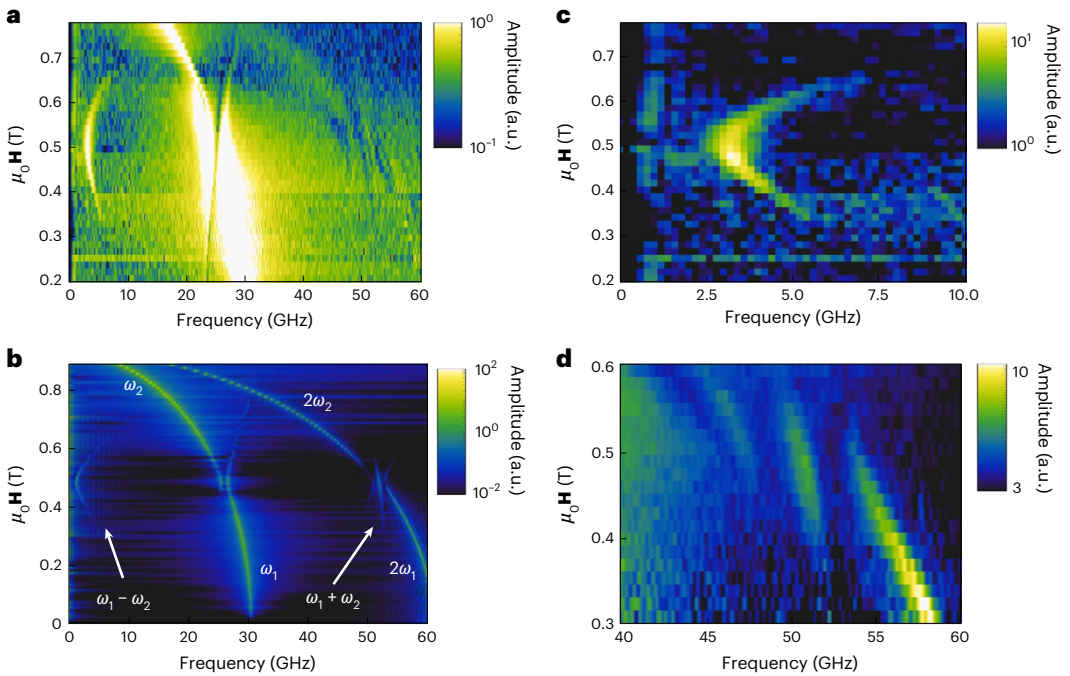

Fig. 2 | Nonlinear coupling of coherently hybridized acoustic and optical magnon modes. a, Measured magnon spectrum with $\theta_{ab} = 2°$. **b**, Simulated magnon spectrum with $\theta_{ab} = 2°$, containing only nonlinear terms. **c**,**d**, Measured magnon spectra with axes scaled to highlight the optical sideband resulting from the exciton coupling to the difference-frequency (**c**) and sum-frequency (**d**) generation between the hybridized magnon modes.

fluence-dependent effects. The fluence dependences in Fig. 1f,g confirm the nature of the $2\omega_0$ mode as SHG, where strong interactions of pump-generated magnons push the system beyond linear spin-wave theory.

This SHG is a manifestation of the nonlinearity of the spin-wave dynamics that can be described by the Landau–Lifshitz–Gilbert (LLG) equations, which are inherently nonlinear. These dynamics, and the power dependence of the SHG mode, are indicative of perturbative nonlinearity. Perturbative expansions can produce a few harmonics with rapidly decaying intensity. In contrast, in optical HHG it is well known that higher-order harmonics scale non-perturbatively, as they arise from different mechanisms[27–29]. As we will show later in this report, there is also non-perturbative magnon harmonic generation at play that produces extreme HHG. This magnon HHG does not follow the picture presented in Fig. 1c or the perturbative expansion of the LLG equations.

Non-degenerate magnon frequency mixing

We next seek to control the nonlinear response by exploiting the anisotropy of the system. Recent reports[8,25] have demonstrated that off-axis magnetic fields can break C_2 rotational symmetry in CrSBr, hybridizing the optical and acoustic magnons and resulting in two magnon modes that couple to the exciton. Specifically, applying fields at an angle θ_{ab} in the plane of the crystal results in a splitting linearly proportional to θ_{ab} for angles smaller than ~15° (see Methods for simulation details)[8]. Therefore, we expect that nonlinear frequency mixing between the two modes will produce magnon sidebands beyond SHG. We perform the same transient optical reflectivity experiments as described above, but with $\mu_0\mathbf{H}$ applied in the *ab* plane at $\theta_{ab} = 2°$.

Figure 2a shows the magnon spectrum resulting from the FFT of the transient optical measurements. For comparison, Fig. 2b is a simulated spectrum based on the LLG equation, where ω_1 and ω_2 represent the two magnon modes originating from the hybridization of the acoustic and optical magnon modes and their relative spectral weight depends on the contribution of the optical magnon to each hybridized mode. In both the measured and simulated spectra, two features are visible that are not integer multiples of either ω_1 or ω_2 but are their linear combinations (see Fig. 2c,d for a zoomed-in view of the data highlighting these peaks). At $\omega_1 + \omega_2$, there is a mode with weak field dependence sitting directly in the centre of the avoided crossing of the SHG modes which corresponds to the SFG of the magnon modes. Close inspection of Fig. 2d appears to show an unexpected evolution of the spectral weight distribution with increasing magnetic field. This is caused by an additional change of the exciton resonance energy as the field is increased, resulting in an optical signal from the magnon that appears to have a slightly distorted field dependence. At $\omega_1 - \omega_2$, we see a strongly field-dependent feature whose field dependence switches sign at the crossing point of ω_1 and ω_2, $\mu_0\mathbf{H} \approx 0.5$ T, corresponding to the DFG of the fundamental modes.

We can tune the SFG and DFG through the control of θ_{ab}, which tunes the splitting between the ω_1 and ω_2 modes[8,25]. Using the DFG mode as an example, Fig. 3a shows the magnon spectrum as the angle between the in-plane magnetic field and the intermediate magnetic axis *a*, θ_{ab}, is varied at fixed $\mu_0\mathbf{H} = 0.5$ T (time-domain data in Extended Data Fig. 3). The small splitting between ω_1 and ω_2 seen at $\theta_{ab} = 0°$ is probably due to imperfect sample mounting leading to a small tilt with respect to the in-plane magnetic field. As the splitting of the fundamental modes increases with θ_{ab} (ref. 8), a low-frequency DFG mode appears with its frequency increasing linearly with θ_{ab}. Here the disappearance of the DFG signal at frequencies lower than ~2 GHz is a result of our time-domain background subtraction. We find that as θ_{ab} is further increased the DFG mode can be tuned into resonance with the lower branch (ω_2) of the hybridized modes at $\theta_{ab} \approx 26°$. Extended Data Fig. 4 shows a simulation consistent with the experimental results, as well as measurement of the DFG mode at both positive and negative field angles.

Under the resonant condition, we see a small amplitude increase in the ω_2 mode. This amplitude increase is due to parametric amplification of the ω_2 magnon mode, as has been reported in synthetic AFMs[30]. Figure 3b,c shows energy-level diagrams of two scenarios under which parametric amplification of the ω_2 mode would occur. In Fig. 3b, an ω_1 magnon and an ω_2 magnon, where $\omega_1 > \omega_2$, are present. Here, the ω_2 magnon induces the downconversion of the ω_1 magnon by resonantly

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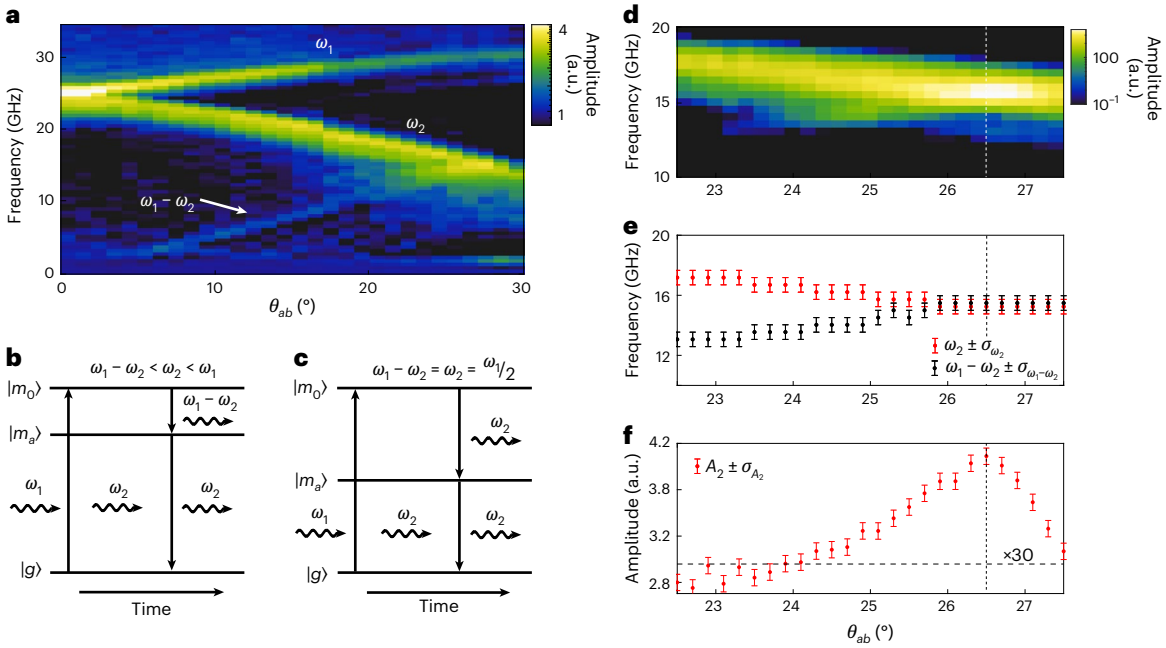

Fig. 3 | Tunable DFG and parametric magnon amplification in CrSBr. **a**, Measured magnon spectrum with respect to magnetic field angle θ_{ab}. **b**, Schematic representation of a parametric downconversion process, where an ω_2 magnon stimulates the downconversion of an ω_1 magnon into two new magnons with distinct energies. **c**, Schematic representation of parametric amplification, where an ω_1 magnon is split by an ω_2 magnon into two additional ω_2 magnons. **d**, Measured magnon spectrum focused on the region of overlap between the DFG and ω_2 magnon modes. **e**, Frequencies of the ω_2 (red) and DFG (black) modes extracted from **d**. The s.d. of the peak locations from 25 measurements was smaller than our pixel size, which is set as an upper bound on the fitting error. **f**, Amplitude of the ω_2 peak extracted from **d**, where error bars signify the s.d. between 25 measurements. The black dashed line indicates the amplitude of the DFG mode (multiplied by 30) when the two peaks are not overlapped.

stimulating the emission of an ω_2 magnon and another magnon at $\omega_1 - \omega_2$. However, the amplification is solely dependent on the ω_1 and ω_2 amplitudes and is not strictly tunable in this regime[30]. However, under the special condition $\omega_1 - \omega_2 = \omega_2$, we can see that the DFG magnon is resonant with ω_2, as shown in Fig. 3c. Here, the downconversion of the ω_1 magnon releases two additional ω_2 magnons, which can then stimulate further downconversion processes.

Figure 3d presents a higher-resolution measurement near the crossing of the DFG and ω_2 modes, where the parametric amplification is evident. The extracted frequencies of the DFG (black dots) and ω_2 (red dots) magnons are shown in Fig. 3e, where the DFG frequencies are calculated from the difference of the ω_1 and ω_2 peaks. Further, Fig. 3f shows the extracted amplitudes of the ω_2 and DFG peaks. We see that there is a distinct amplitude increase, about 1.5-fold amplification, in the ω_2 magnon when the two modes are in resonance, compared with the off-resonant case. We can also compare this increase with the DFG amplitude measured away from their overlap region (black dashed line in Fig. 3f, measured at $\theta_{ab} \approx 22°$ and multiplied by 30). The linear sum of the two modes off resonance is much smaller than the mode amplitude on resonance, further supporting the observation of parametric amplification. The tunability of the nonlinear response offers a fascinating platform for magnon frequency conversion, generation of entangled magnons, and functional magnonic devices based on nonlinear interactions.

Extreme magnon HHG

In addition to the nonlinear physics presented so far, we have found that CrSBr is remarkable for studying extreme magnonic nonlinearities, such as HHG. Figure 4a presents a field-dependent magnon spectrum that shows a series of sidebands at integer multiples of ω_0, ranging from ~5 GHz to over 600 GHz. Figure 4b shows a linecut from this spectrum at a field of $\mu_0\mathbf{H} \approx 0.3$ T, presented in units of ω_0. In these data, an astonishingly high harmonic series can be seen, with the inset of Fig. 4b presenting harmonics beyond the 20th order. The slope of the field-dependent magnon frequency exhibits clear changes across the higher harmonics. To understand this behaviour, we consider a magnon with the fundamental frequency $\omega(B_a)$, and its nth harmonic, with frequency $n\omega(B_a)$. The slope of the fundamental frequency at any field value will be $\frac{\partial\omega(B_a)}{\partial B_a}$ and the slope of the nth harmonic will be $n\frac{\partial\omega(B_a)}{\partial B_a}$, resulting in an increase in the field dependence as the order increases.

The HHG processes seen above are repeatable across samples (additional HHG data in Extended Data Fig. 5). Notably, the higher harmonics show no appreciable increase in linewidth. As mentioned previously, the observed magnon decay is caused primarily by the coherent propagation of the magnons[20]. Thus, the relatively constant linewidths here suggest that the group velocities of the harmonics are not substantially different from that of the fundamental magnons. We have also checked that the harmonic orders here are not artefacts of the FFT or data processing, evidenced by the additional peaks in the avoided crossings of the higher-harmonic bands. These peaks, like the SFG peaks discussed in Fig. 2, are caused by higher-order nonlinear mixing processes. They can be seen in Extended Data Fig. 5 as a blurring of the high-harmonic sidebands around $\mu_0\mathbf{H} \approx 0.5$ T. These peaks, which are not harmonic replicas of the individual bands but mixing processes, require nonlinear interaction in the system and confirm that the observed HHG is a real effect.

Until very recently, only the first few magnon harmonic orders were measurable[11,13,14] using conventional techniques such as Brillouin light scattering and FM resonance spectroscopies. Recently, Fabry–Perot cavities[31] have achieved up to 17th harmonic generation and resonant terahertz experiments have generated up to sixth order[22]. To our knowledge, one report has employed electron spin resonance of NV centres in nanodiamonds[32] to measure harmonic orders (>50) exceeding those presented here. However, the detection scheme employed in that study limited the frequencies attainable to ~3 GHz and accessed the continuous spin-wave dispersion regime. In our

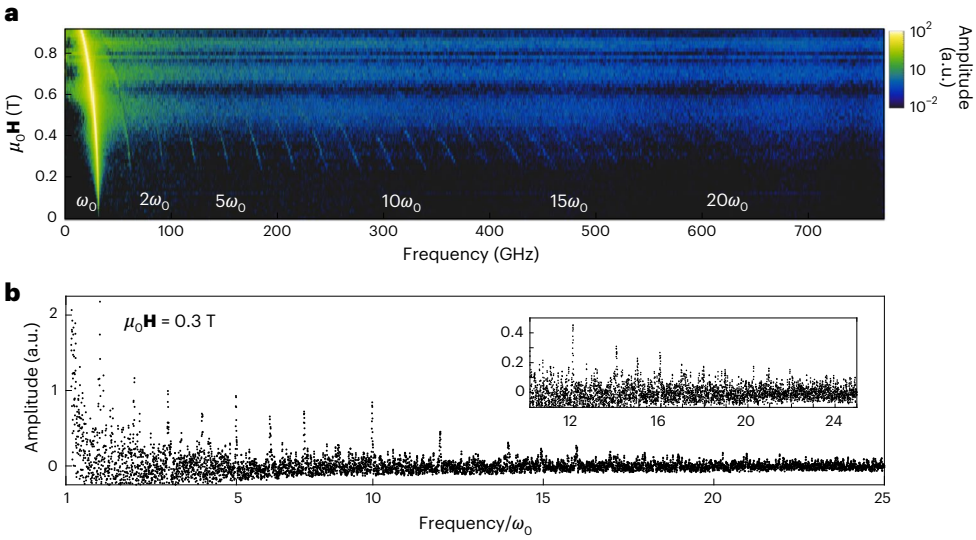

Fig. 4 | Magnon high-harmonic generation. a, Field-dependent magnon spectrum presenting high-harmonic magnon modes. **b**, Linecut taken from **a** at μ_0**H** ≈ 0.3 T, presenting harmonics beyond 20th order. Inset: harmonics of 11th order and higher with the axes rescaled. In this figure, the fundamental magnon extends past the vertical scale to an amplitude of ~100.

experiments, we generate dozens of harmonics of a discrete magnon and the frequency window is only physically limited by the time duration of our laser pulses, which could extend well into the terahertz range. The high sensitivity of the optical measurements, enabled by the exciton–magnon coupling, allows us to use weak excitation pulses (<300 μJ cm^{-2}). We expect that the use of amplified laser systems or engineered magnon cavities will more effectively access the nonlinearity, which could push the HHG further into the terahertz regime. This could allow coherent control of the magnetization and coupling to the known A^1_{1g} phonon mode at ~3.6 THz (ref. 33). The coupling of excitons and nonlinear magnons leads to exciting possibilities in quantum transduction, expanding the range of accessible frequencies across hundreds of gigahertz. This could allow a wide range of low-energy excitations to be coupled to optical photons and measured with standard optical detectors.

The observed harmonic series in Fig. 4 is the magnonic analogue of optical HHG processes seen in atomic gases[27] and solids[28] that are highly useful for frequency conversion, measurement of crystal symmetries[34] and probing topological order[35]. This HHG indicates a deviation from the perturbative expansion of the LLG equations described earlier into the non-perturbative nonlinear regime. In contrast to the perturbative nonlinearity where the amplitude of high harmonics decays exponentially as the order increases, our results in Fig. 4b show that the amplitude presents a plateau after a decay in the first few orders, serving as definitive evidence for the non-perturbative nature of the HHG. This is similar to non-perturbative optical HHG, where a plateau in the amplitude also forms beyond the first few orders[36]. However, in contrast to optical HHG in atomic gases, which show only odd harmonics, we see both even and odd harmonics, and the magnetic field does not affect which harmonics are generated.

The observation of even-order harmonics underscores the unique nature of magnonic HHG in comparison to optical HHG, where only odd-order HHG is seen. In optical HHG, the process originates from electrons moving in a nonlinear potential depending on the electron coordinate x. Due to the inversion symmetry of the atoms, this nonlinearity must be an even order of x, leading to the absence of even-order harmonics. While the mechanism that generates the magnon HHG is still unknown, we have developed a phenomenological model that accurately recreates the HHG arising from the nonlinear dynamics of the quasiparticles characterized by a generalized coordinate Q in configuration space. This Q can be even under the symmetry operations of solid media, allowing for both even- and odd-order nonlinearities in the potential energy function, which enables the generation of both odd- and even-order harmonics. The difference in amplitude between odd and even harmonics may reflect the relative strengths of the corresponding nonlinear terms for these quasiparticles. This nonlinear oscillator could be a higher-energy magnon or phonon mode in the system. Extended Data Fig. 6 shows a simulated spectrum, which qualitatively matches the experimental data (Methods).

Conclusions

Nonlinear magnonics holds the potential to become an important component in information systems that are smaller, more energy efficient and more robust[37] than those based on electronic circuits[4]. We have shown that CrSBr is an excellent candidate, as it displays a strongly nonlinear magnon response in the microwave range necessary for electronic measurement. The demonstrated unique optical access to the harmonic orders of the magnons facilitates magnon resonances over a much greater range and extends CrSBr's potential as a quantum transducer[8,38]. In addition, the SFG and DFG magnons provide avenues for generating entangled magnons[39] and parametric amplification of weak magnon signals[30] with unprecedented tunability via the angle of an applied magnetic field. Finally, the observed modulated exciton states from the extreme nonlinear magnonic processes provide access to optomagnonic physics and could open possibilities for magnonic control of the excitonic response, or vice versa. The nonlinear response presented here, combined with the desirable energy range and optical coupling, further cements CrSBr as a promising candidate for implementing hybrid magnon information processing.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41565-025-01890-8.

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Methods

Crystal growth

The following reagents were used as received unless otherwise stated: chromium powder (99.94%, −200 mesh, Alfa Aesar), sulfur pieces (99.9995%, Alfa Aesar), bromine (99.99%, Aldrich) and chromium dichloride (anhydrous, 99.9%, Strem Chemicals). To begin, $CrBr_3$ was synthesized from Cr (1.78 g, 34.2 mmol) and Br_2 (8.41 g, 52.6 mmol) with one end of the tube held at 1,000 °C and the other end at 50 °C with a water bath. Of note, one end of the tube must be maintained below 120 °C to prevent the tube from exploding from bromine overpressure. Chromium (0.174 g, 3.35 mmol), sulfur (0.196 g, 6.11 mmol) and $CrBr_3$ (0.803 g, 2.75 mmol) were loaded into a 12.7-mm-outer-diameter, 10.5-mm-inner-diameter fused silica tube. The tube was evacuated to ~30 mTorr and flame sealed to a length of 20 cm. It was then was placed into a computer-controlled, two-zone, tube furnace. The source side was heated to 850 °C over 24 h, allowed to soak for 24 h, heated to 950 °C over 12 h, allowed to soak for 48 h and then cooled to ambient temperature over 6 h. The sink side was heated to 950 °C over 24 h, allowed to soak for 24 h, heated to 850 °C over 12 h, allowed to soak for 48 h and then cooled to ambient temperature over 6 h. The crystals were cleaned by soaking in a 1 mg ml^{-1} $CrCl_2$ aqueous solution for 1 h at ambient temperature. After soaking, the solution was decanted and the crystals were thoroughly rinsed with deionized water and acetone. Residual sulfur residue was removed by washing with warm toluene.

Sample fabrication

CrSBr crystals were exfoliated onto silicon wafers with a 90 nm silicon dioxide layer and thin (tens to hundreds of nanometres) flakes were identified by optical contrast. Though thin bulk flakes were used, the magnon has weak thickness dependence[7] and we saw no evidence of thickness-dependent nonlinear effects in our measurements.

Optical measurements

Transient optical reflectivity measurements were performed by tuning the output of a titanium sapphire oscillator to 1.33 eV, overlapping with one of the CrSBr exciton states. The output was frequency doubled and the second harmonic and fundamental were separated into pump and probe arms of the experiment by a dichroic mirror. The probe beam was sent to a retroreflector mounted on a motorized translation stage to produce the pump–probe delay. Each beam was sent through a waveplate and polarizer to simultaneously attenuate the beams and align their polarization to the crystal axes. The beams were recombined and sent through a 0.6 numerical aperture microscope objective onto the sample. The back-reflected beam was measured on a photodiode with a lock-in amplifier demodulating the signal at the frequency of a mechanical chopper placed in the pump arm of the experiment. To produce the time-domain data, the delay stage was continuously swept at low speed while streaming data from the lock-in amplifier to the host computer at a high sampling rate (>100 kHz), which produced time traces with 400 fs resolution in the data presented here. Multiple traces ($4 < N < 25$) were recorded and averaged as well as being binned in the time domain, depending on the desired signal-to-noise ratio and frequency range of the spectra. The error bars in Fig. 3 are the s.d. of the extracted values over all N measurements, while the centre is the average value. The samples were kept at 35 K, far below the Néel temperature of ~132 K (ref. 24), in an optical cryostat with an integrated vector magnet capable of applying fields up to 1 T along any arbitrary direction on the unit sphere. To determine the orientation of our applied field with respect to the crystal axes, a separate measurement was made with the field amplitude set to the centre of the avoided crossing while we varied only the field angle. Zero angle was then set to be the point where the splitting, Δ, went to zero. Field angles were then set by appropriately varying the two orthogonal in-plane magnetic fields in our vector magnet.

Simulation of nonlinear magnon spectra

CrSBr is a layered material with ordered spin where the intralayer coupling is FM while the interlayer coupling is AFM. In each layer, there are two magnetic atoms in the unit cell that form a square lattice. The nearest-neighbouring magnetic atoms in two layers have a relative in-plane shift. In principle, we need to consider two magnetic atoms per layer per unit cell and the exchange coupling between second- or third-nearest-neighbouring sites may play a role. However, for simplicity, we consider only one magnetic atom per layer per unit cell and consider only the nearest-neighbouring exchange interaction. The validity of this simplified model is justified by previous comparison with experimental results[8].

The Hamiltonian reads

$$H = J_2\mathbf{S}_1\cdot\mathbf{S}_2 - g\mu_B\mathbf{H}\cdot\sum_{i=1,2}\mathbf{S}_i + \sum_{i=1,2} -K_b\left(\mathbf{S}_i\cdot\hat{\mathbf{b}}\right)^2 + K_c(\mathbf{S}_i\cdot\hat{\mathbf{c}})^2,$$

where $\mathbf{S}_{1,2}$ denotes the spin in layers 1 and 2, with $J_2 > 0$ for AFM interlayer coupling. The easy and hard axes are the b and c axes, respectively, with $K_{b,c} > 0$. $\mathbf{H}$ is the external magnetic field, with g and μ_B denoting the g factor and the Bohr magneton. In our simulations, the values used in the Hamiltonian are $S = 3/2$, $g = 2$, $J_2 = -0.0224$ meV, $K_b = 0.0113$ meV and $K_c = 0.0355$ meV.

Here, we solve the full LLG equation written as

$$\dot{\mathbf{S}}_1 = -\mathbf{S}_1\times\mathbf{H}_1$$
$$\dot{\mathbf{S}}_2 = -\mathbf{S}_2\times\mathbf{H}_2,$$

with effective magnetic fields

$$\mathbf{H}_1 = J_2\mathbf{S}_2 - g\mu_B\mathbf{H} - 2K_b\mathbf{S}_1\cdot\hat{\mathbf{b}}\hat{\mathbf{b}} + 2K_c\mathbf{S}_1\cdot\hat{\mathbf{c}}\hat{\mathbf{c}}$$
$$\mathbf{H}_2 = J_2\mathbf{S}_1 - g\mu_B\mathbf{H} - 2K_b\mathbf{S}_2\cdot\hat{\mathbf{b}}\hat{\mathbf{b}} + 2K_c\mathbf{S}_2\cdot\hat{\mathbf{c}}\hat{\mathbf{c}}.$$

In the presence of a magnetic field along the a direction, the spins tilt along the field direction and the tilting angle can be determined by $\partial H/\partial\phi = 0$. Here, the static H is

$$\begin{aligned} H(\phi) &= J_2S^2\cos(\pi - 2\phi) - 2g\mu_BSH_a\sin\phi - 2K_bS^2\cos^2\phi \\ &= -J_2S^2\cos(2\phi) - 2g\mu_BSH_a\sin\phi - 2K_bS^2\cos^2\phi \end{aligned}$$

and

$$\begin{aligned} \frac{dH}{d\phi} &= 2J_2S^2\sin(2\phi) - 2g\mu_BSH_a\cos\phi + 4K_bS^2\cos\phi\sin\phi \\ &= 2(J_2 + K_b)S^2\sin(2\phi) - 2g\mu_BSH_a\cos\phi. \end{aligned}$$

Thus,

$$\sin\phi = \frac{g\mu_BH_a}{2(J_2+K_b)S}.$$

When the magnetic field deviates from the a direction, the tilting angles for spin 1 and spin 2 are different. Nevertheless, we can still find the equilibrium spin configuration by minimizing the total energy with respect to the tilting angles.

After the arrival of the pump pulse, the equilibrium spin configuration changes due to the transient excited electronic states. We model this effect by introducing a sudden change in the spin configuration from its equilibrium orientation by a small amount. When the magnetic field is along the a direction, the system respects two-fold rotational symmetry about the a axis. This guarantees a symmetric configuration after the arrival of the pump pulse. Therefore, our introduced change is chosen to respect this symmetry. When the magnetic field deviates from the a axis slightly, this symmetry still approximately holds. We then numerically solve the LLG equation to obtain $\mathbf{S}_{1,2}(t)$ as a function of time, t. $\mathbf{S}_1(t)\cdot\mathbf{S}_2(t)$, whose change is proportional to the shift in

the exciton energy, can then be obtained. We then Fourier transform $\mathbf{S}_1(t) \cdot \mathbf{S}_2(t)$ to obtain the full spectrum of the magnon modes.

We introduce a phenomenological model to explain the HHG observed in our data. This model consists of an anharmonic oscillator that is bilinearly coupled to the coherently excited magnon. The physical nature of the anharmonic oscillator will be discussed later. It is described by the following Hamiltonian:

$$H = \frac{P^2}{2} + \frac{1}{2}\omega^2 Q^2 + \frac{\upsilon_3}{3}Q^3 + \frac{\upsilon_4}{4}Q^4,$$

where P and Q are the generalized momentum and coordinate of the oscillator, ω is the intrinsic frequency of the oscillator and $\upsilon_{3,4}$ represent the nonlinear coupling coefficients within the oscillator. This nonlinearity differs from that in optical HHG. Models for optical HHG study the electron dynamics in real space, where the generalized coordinate represents the real-space coordinate that is odd under inversion symmetry. The inversion symmetry of the atomic potential ensures that the cubic-order nonlinearity vanishes. However, in our model shown above, Q corresponds to the generalized coordinate for the normal mode of an elementary excitation that can be even under symmetry operations such as inversion or two-fold rotation about the a axis. Consequently, the symmetry permits both cubic- and quartic-order nonlinearities. This crucial difference enables our observation of even-order harmonics, which are absent from optical HHG.

The presence of coherent optical magnons, which exhibit strong oscillation over the ~3 ns measurement period, generate a periodic driving force, $F(t)$, on the anharmonic oscillator. Its equation of motion reads

$$\ddot{Q} = -\omega^2 Q - \upsilon_3 Q^2 - \upsilon_4 Q^3 - \frac{\dot{Q}}{\Gamma} - F(t)$$

where Γ is the damping coefficient and $F(t)$ oscillates at the optical magnon frequency ω_0. When $\upsilon_3 = 0$, this equation reduces to the well-known Duffing equation, a canonical model for studying nonlinear physics. Although the magnons exhibit damping, leading to an exponential decay of the driving force, experiments have shown that the magnon lifetime exceeds the measurement period[20]. Therefore, we assume a periodic driving force. The results are qualitatively the same with weak decay.

By numerically solving the above equation, we find that the motion of the anharmonic oscillator exhibits high harmonics of the driving frequency ω_0. As shown in Extended Data Fig. 6, the oscillator's dynamics display frequencies that are integer multiples of the driving frequency, including both even and odd harmonics up to approximately the 22nd order. The amplitudes of the Fourier components initially decrease, then plateau, and subsequently decrease again, which qualitatively agrees with the measured data. The linewidths of the peaks show a weak dependence on the harmonic order, also in agreement with experimental observations. The model reliably generates HHG phenomena across a wide range of parameters, provided that nonlinearity and driving are present. We note that a non-zero υ_3 is essential in generating even harmonics. By tuning the relative amplitudes of υ_3 and υ_4, we can control the relative amplitudes of even- and odd-order harmonics, which are related to the even–odd oscillation of the amplitudes of the high harmonics in Fig. 4b.

This phenomenological explanation relies on the assumptions that (1) the anharmonic oscillator is bilinearly coupled to the optical magnon and (2) the anharmonic oscillator also couples to the exciton. Since the ultrafast pump pulse excites electronic excitations instantaneously, it is expected to suddenly modify the spin interactions and anisotropy, enabling a bilinear coupling between the optical magnon and other high-frequency magnons. These magnon modes can also couple to the exciton as they alter the relative spin angles. Alternatively, the anharmonic oscillator could also represent phonons, such as interlayer shear phonons, which can modify spin-exchange pathways and thus couple to magnons. These phonons can exhibit nonlinearity and couple to excitons as well. It is possible that both processes contribute to the HHG.

While the phenomenological model produces results that qualitatively agree with our experiments, the microscopic mechanism remains elusive. Moreover, there are alternative mechanisms for generating HHG, such as inhomogeneity in soft FM thin films with dipole–dipole interaction under a periodic driving. To elucidate the microscopic origin of the HHG quantitatively, further experiments and theoretical calculations are required, which we leave for future work.

Data availability

The datasets generated during and/or analysed during this study are provided with this paper. All other data that support the plots within this paper and other findings of this study are available from the corresponding author upon reasonable request. Source data are provided with this paper.

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Author contributions

X.X., D.X., Y.R. and G.M.D. conceived the project. G.M.D. performed the measurements with help from M.N., S.P., J.C. and J.F. M.N., S.P. and J.C. fabricated the samples. G.M.D., M.N., J.C., Y.J.B., X.Z., Y.R., D.X. and X.X. analysed the data and interpreted the results. Y.R. and D.X. built the model and performed the simulations. D.G.C. and X.R. grew the CrSBr crystals. G.M.D., M.N., X.Z., Y.R., D.X. and X.X. wrote the manuscript with input from all authors. All authors discussed the results.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41565-025-01890-8.

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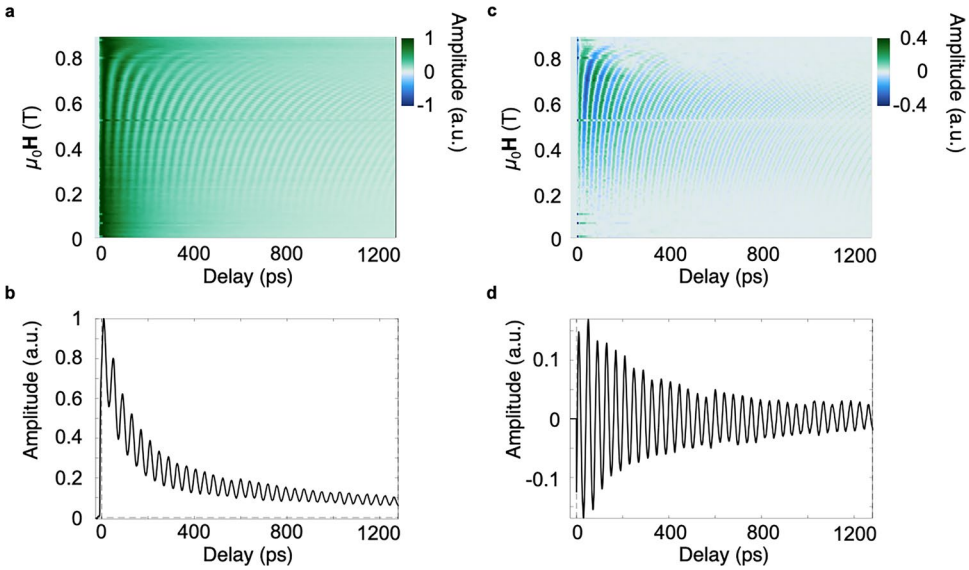

Extended Data Fig. 1 | Field dependent transient reflectivity data. a, Raw transient reflectivity data corresponding to the spectrum presented in Fig. 1d. **b**, Linecut taken from (a) at $\mu_o H \approx 0.45$ T. **c**, Same data as in (a) with the exponential decay removed. **d**, Linecut taken from (c) at $\mu_o H \approx 0.45$ T.

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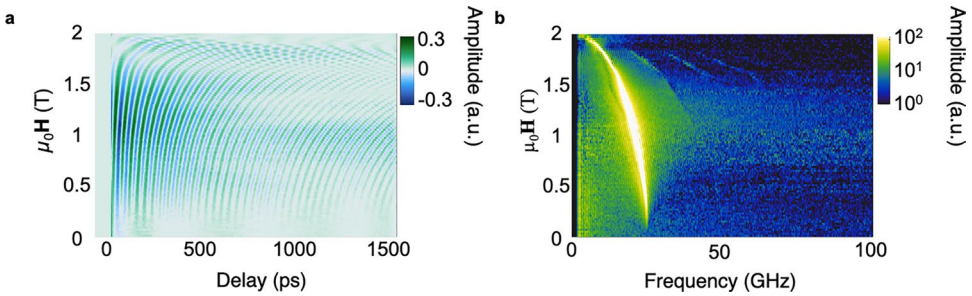

Extended Data Fig. 2 | Magnon HHG under *c* axis field. a, Transient reflectivity data from CrSBr under a field applied along the *c* (hard) crystal (magnetic) axis. **b**, Corresponding magnon spectrum presenting the first few harmonic orders.

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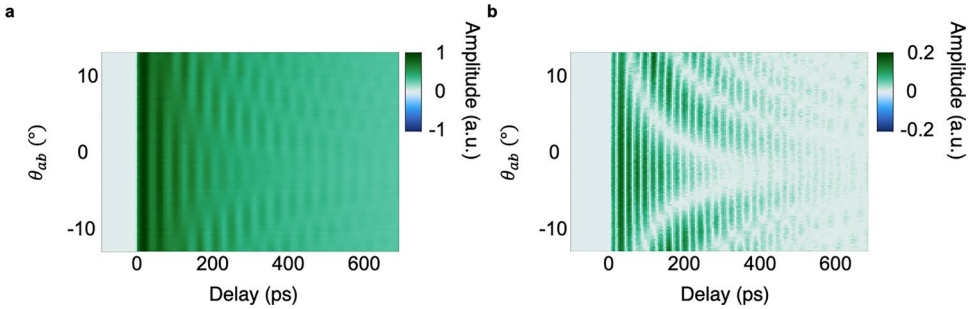

Extended Data Fig. 3 | Field angle dependent transient reflectivity data. a, Raw transient optical reflectivity data corresponding to the spectrum presented in Fig. 3a. **b**, Same data as in (a) with the exponential decay removed.

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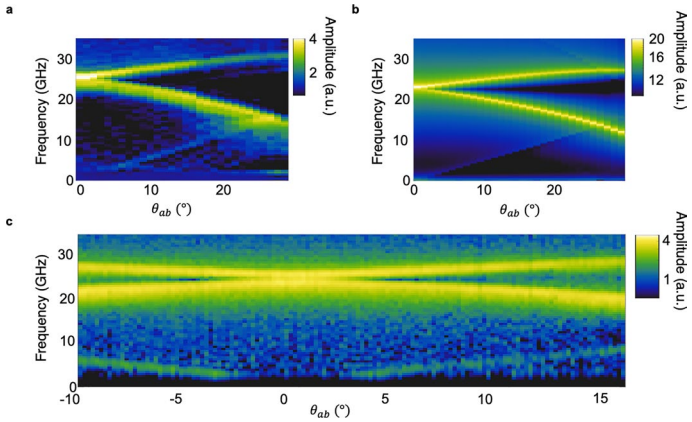

Extended Data Fig. 4 | Comparison of field angle dependent data with simulations. a, Same measured magnon spectrum presented in Fig. 3b. **b**, Simulated magnon spectrum. **c**, Additional dataset showing the DFG mode at positive and negative field angles.

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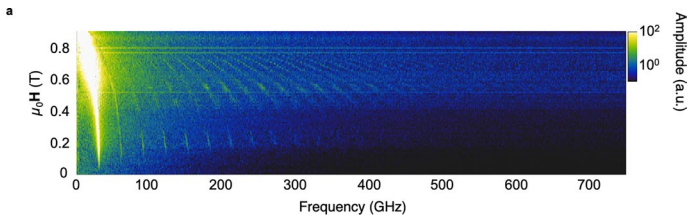

Extended Data Fig. 5 | HHG spectrum from an additional CrSBr sample. a, Magnon spectrum presenting HHG from a ~ 90 nm thick flake, roughly half the thickness of the sample shown in the main text.

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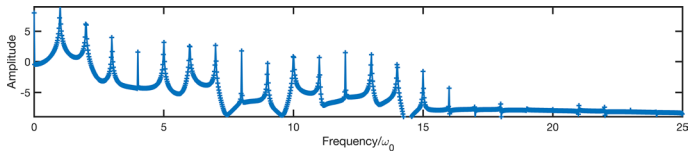

Extended Data Fig. 6 | Model calculation of HHG. The amplitude of the Fourier components as a function of frequency is shown. The amplitude first decays, then reaches a plateau, and then decays again. Both even and odd harmonics exist. The model parameters used to produce this spectrum are $\omega = 1 = 10\omega_0$, $\upsilon_3 = 1$, $\upsilon_4 = \frac{1}{2}$, $\Gamma = 10$, and $F(t) = 0.5\sin(\omega_0 t)$.